



**ABANTU
ENVIRONMENTAL
SERVICES**



078 207 8278



info@abantuenvironmental.co.za

Fax : 086 685 9536

PROJECT NAME:

PROPOSED CONSTRUCTION OF DUTYWA CONFERENCE CENTRE AND ASSOCIATED INFRASTRUCTURE WITHIN THE MBHASHE LOCAL MUNICIPALITY, IN THE EASTERN CAPE.

REPORT TITLE:

AQUATIC BIODIVERSITY IMPACT ASSESSMENT

DEDEAT REFERENCE: EC/9/A/LN1, LN3/M/26-23

DATE: AUGUST 2026

ENVIRONMENTAL ASSESSMENT PRACTITIONER:

ABANTU ENVIRONMENTAL SERVICES (PTY) LTD



**ABANTU
ENVIRONMENTAL
SERVICES**

CONSULTING ENGINEER:

CHIMORA GROUP (PTY) LTD



ON BEHALF OF:

MBHASHE LOCAL MUNICIPALITY



Mbhashe
Local Municipality
WILLOWVALE | DUTYWA | ELLIOTDALE

OFFICE	ADDRESS	CONTACT DETAILS
Gqeberha	41 Avon Place Overbaakens Gqeberha 6070	Contact Person: Mr Sive Mlamla (Pr.Sci.Nat, Reg EAP) Cell: 078 207 8278 Phone: 041 582 2648 Fax: 086 685 9536 Email: info@abantuenvironmental.co.za Website: www.abantuenvironmental.co.za

DOCUMENT REVISION STATUS

REVISION	PREPARED BY:	ASSISTED BY:
00	Mandilakhe Msuthu 	Ms Zimkita Dalasile 
DATE	August 2026	August 2026

DECLARATION OF INDEPENDENCE

I, **Mandilakhe Msuthu** , in my capacity as a specialist consultant, hereby declare that I -

- Act as an independent consultant;
- Do not have any financial interest in the undertaking of the activity, other than remuneration for the work performed in terms of the National Environmental Management Act, 1998 (Act 107 of 1998);
- Have and will not have vested interest in the operations of the facility;
- Have no, and will not engage in, conflicting interests in the undertaking of the activity;
- Undertake to disclose, to the competent authority, any material information that has or may have the potential to influence the decision of the competent authority or the objectivity of any report, plan or document required in terms of the National Environmental Management Act, 1998 (Act 107 of 1998);
- Will provide the competent authority with access to all information at my disposal regarding the application, whether such information is favourable to the applicant or not;
- As a registered member of the South African Council for Natural Scientific Professions, will undertake my profession in accordance with the Code of Conduct of the Council, as well as any other societies to which I am a member;
- Based on information provided to me by the project engineer and proponent, and in addition to information obtained during the course of this study, have presented the results and conclusion within the associated document to the best of my professional ability; and
- Undertake to have my work peer reviewed on a regular basis by a competent specialist in the field of study for which I am registered.



Mr Mandilakhe Msutu (Pr.Sci.Nat)
Aquatic Biodiversity Specialist

CONTACT INFORMATION AND EXPERTISE

Table 1. Environmental Specialist details

EAP/Specialist:	Mr Mandilakhe Msutu (Pr. Sci.Nat)
Expertise:	Mandillakhe Msutu is an environmental scientist with 10 years of experience environmental consulting and scientific research. His expertise include: • Environmental monitoring and environmental compliance audits. • Aquatic Biodiversity Assessments • Water quality and Air Quality monitoring • Bio-monitoring on rivers (including SASS) • Environmental Impact Assessments Qualifications • BSc Hons in Environmental and Water Science • BSc Environmental and Water Science Registrations and memberships: • Professional Natural Scientist (Pr. Sci.Nat) (SACNASP) Registration No. 116842

Table 2. Assistant specialist

Assistant Specialist:	Ms Zimkita Dalasile (Cand.Sci.Nat (SACNASP)
Expertise:	Junior Environmental Scientist with 4 years' experience in environmental consulting and scientific research. My expertise includes environmental management, water resource management, water quality management, aquatic ecosystems, environmental impact assessments, scientific report writing, data analysis, environmental monitoring and environmental compliance audits. Qualifications • MSc Environmental Management with Ecological Water Requirements (Management of Ecological Drivers in Aquatic Systems & Management of Ecological Responders in Aquatic Systems) Mini Dissertation: Strategies to manage nutrient inputs and their effectiveness: A case study of the Hartbeespoort Dam • Postgraduate Diploma Integrated Water Resource Management (Water Resource Management, Spatial Analysis for IWRM, Environmental Management & Water Regulations, Water Quality Management and Aquatics Ecosystems Management) • BSc. Environmental and Water Science (Geology, Environmental Assessment and Management, Geohydrology, Hydrology and Advanced Geographical Information Systems for Environmental System) Professional registration: • Candidate Natural Scientist (Cand.Sci.Nat) Registration No. 132501 Training: • Introduction Environmental Impact Assessments (EIA) (2024)

EXECUTIVE SUMMARY

The Mbhashe Local Municipality is proposing the construction of a conference centre and associated infrastructure in Dutywa, within the Mbhashe Local Municipality, Eastern Cape. The study was commissioned to inform the Environmental Authorisation process and to assess the potential impacts of the proposed development on rivers, drainage lines, riparian habitat and associated aquatic biodiversity. The assessment was undertaken in accordance with the Environmental Impact Assessment Regulations, 2014 (as amended) and the Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Aquatic Biodiversity (GN 320 of 2020).

The assessment included a desktop review of available aquatic biodiversity information and a once-off field survey undertaken during July 2026. The field assessment confirmed the presence of the Mputi River, which is mapped as a non-perennial watercourse. Four sites were assessed along the river, extending from an upstream section to the downstream confluence. The assessed reach occurs within a Lower Foothill geomorphic zone and was generally characterised by a narrow and shallow channel, with low-flow conditions and slow to moderate surface flow observed during the survey. Field observations indicated that the Mputi River is subject to existing anthropogenic pressures, including localised bank erosion, sediment deposition, channel disturbance associated with existing crossings and infrastructure, stormwater-related inputs and solid-waste accumulation within the channel and riparian vegetation. Despite these pressures, portions of the river retain relatively well-established riparian vegetation.

The Index of Habitat Integrity (IHI) assessment classified the instream habitat as Integrity Class C (Moderately Modified), while the riparian habitat was classified as Integrity Class B (Largely Natural with Few Modifications). Instream modification was primarily associated with abstraction and flow-related pressures, localised bed and channel modification and solid-waste accumulation. Riparian disturbance was comparatively lower and primarily associated with localised vegetation disturbance, bank erosion and channel modification. The Ecological Importance and Sensitivity (EIS) assessment resulted in a median score of 1.5, corresponding to a Moderate EIS category.

The construction phase impacts identified include loss and disturbance of riparian vegetation and associated habitat, disturbance and modification of the riverbed and banks, bank erosion and sedimentation, temporary flow modification, potential deterioration of surface-water quality and cumulative impacts on aquatic biodiversity. During operation, the impacts include increased stormwater runoff, potential leakage or failure

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of the sewer pipeline and ongoing disturbance of riparian habitat. With implementation of the recommended mitigation measures, most impacts are expected to reduce to low or very low significance.

The proposed development is considered acceptable from an aquatic biodiversity perspective, provided that the recommended mitigation measures are fully implemented.

REPORT REQUIREMENTS

Specialist Report Requirement Checklist (as per gazetted procedures for the assessment and minimum criteria for reporting on identified environmental themes in terms of sections 24(5)(a) and (h) and 44 of the National Environmental Management Act, 1998.)

Table 3. Report requirements

Item No.	Requirement	Report Reference
2.1	The assessment must be prepared by a specialist registered with the South African Council for Natural Scientific Professionals (SACNASP), with expertise in the field of aquatic sciences.	Section 0
2.2	The assessment must be undertaken on the preferred site and within the proposed development footprint.	Section 1.1
2.3	The assessment must provide a baseline description of the site which includes, as a minimum, the following aspects:	Section 3
2.3.1	a description of the aquatic biodiversity and ecosystems on the site, including; (a) aquatic ecosystem types; and (b) presence of aquatic species, and composition of aquatic species communities, their habitat, distribution and movement patterns;	Section 3
2.3.2	the threat status of the ecosystem and species as identified by the screening tool;	Section 1.2
2.3.3	an indication of the national and provincial priority status of the aquatic ecosystem, including a description of the criteria for the given status (i.e. if the site includes a wetland or a river freshwater ecosystem priority area or sub catchment, a strategic water source area, a priority estuary, whether or not they are free-flowing rivers, wetland clusters, a critical biodiversity or ecologically sensitivity area); and	Section 3
2.3.4	a description of the ecological importance and sensitivity of the aquatic ecosystem including: (a) the description (spatially, if possible) of the ecosystem processes that operate in relation to the aquatic ecosystems on and immediately adjacent to the site (e.g. movement of surface and subsurface water, recharge, discharge, sediment transport, etc.); and (b) the historic ecological condition (reference) as well as present ecological state of rivers (in-stream, riparian and floodplain habitat), wetlands and/or estuaries in terms of possible changes to the channel and flow regime (surface and groundwater).	Section 4.2.1 and 4.2.2

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2.4	The assessment must identify alternative development footprints within the preferred site which would be of a “low” sensitivity as identified by the screening tool and verified through the site sensitivity verification and which were not considered appropriate.	N/A
2.5	Related to impacts, a detailed assessment of the potential impacts of the proposed development on the following aspects must be undertaken to answer the following questions:	Section 5
2.5.1	is the proposed development consistent with maintaining the priority aquatic ecosystem in its current state and according to the stated goal?	Section 5
2.5.2	is the proposed development consistent with maintaining the resource quality objectives for the aquatic ecosystems present?	
2.5.3	how will the proposed development impact on fixed and dynamic ecological processes that operate within or across the site? This must include: (a) impacts on hydrological functioning at a landscape level and across the site which can arise from changes to flood regimes (e.g. suppression of floods, loss of flood attenuation capacity, unseasonal flooding or destruction of floodplain processes); (b) will the proposed development change the sediment regime of the aquatic ecosystem and its sub-catchment (e.g. sand movement, meandering river mouth or estuary, flooding or sedimentation patterns); (c) what will the extent of the modification in relation to the overall aquatic ecosystem be (e.g. at the source, upstream or downstream portion, in the temporary / seasonal / permanent zone of a wetland, in the riparian zone or within the channel of a watercourse, etc.); and (d) to what extent will the risks associated with water uses and related activities change;	Section 5.1
2.5.4	how will the proposed development impact on the functioning of the aquatic feature? This must include:	Section 5
	(a) base flows (e.g. too little or too much water in terms of characteristics and requirements of the system); (b) quantity of water including change in the hydrological regime or hydroperiod of the aquatic ecosystem (e.g. seasonal to temporary or permanent; impact of over-abstraction or instream or off-stream impoundment of a wetland or river); (c) change in the hydrogeomorphic typing of the aquatic ecosystem (e.g. change from an unchannelled valley-bottom wetland to a channelled valley-bottom wetland); (d) quality of water (e.g. due to increased sediment load, contamination by chemical and/or organic effluent, and/or eutrophication); (e) fragmentation (e.g. road or pipeline crossing a wetland) and loss of ecological connectivity (lateral and longitudinal); and (f) the loss or degradation of all or part of any unique or important features associated with or within the aquatic ecosystem (e.g. waterfalls, springs, oxbow lakes, meandering or braided channels, peat soils, etc.);	Section 5

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2.5.5	how will the proposed development impact on key ecosystems regulating and supporting services especially: (a) flood attenuation; (b) streamflow regulation; (c) sediment trapping; (d) phosphate assimilation; (e) nitrate assimilation; (f) toxicant assimilation; (g) erosion control; and (h) carbon storage?	
2.5.6	how will the proposed development impact community composition (numbers and density of species) and integrity (condition, viability, predator prey ratios, dispersal rates, etc.) of the faunal and vegetation communities inhabiting the site?	
2.6	In addition to the above, where applicable, impacts to the frequency of estuary mouth closure should be considered, in relation to:	N/A
	(a) size of the estuary; (b) availability of sediment; (c) wave action in the mouth; (d) protection of the mouth; (e) beach slope; (f) volume of mean annual runoff; and (g) extent of saline intrusion (especially relevant to permanently open systems).	N/A
2.7	The findings of the specialist assessment must be written up in an Aquatic Biodiversity Specialist Assessment Report that contains, as a minimum, the following information:	
2.7.1	contact details of the specialist, their SACNASP registration number, their field of expertise and a curriculum vitae;	Appendix A
2.7.2	a signed statement of independence by the specialist;	
2.7.3	a statement on the duration, date and season of the site inspection and the relevance of the season to the outcome of the assessment;	Section 4
2.7.4	the methodology used to undertake the site inspection and the specialist assessment, including equipment and modelling used, where relevant;	Section 2
2.7.5	a description of the assumptions made, any uncertainties or gaps in knowledge or data;	Section 1.3
2.7.6	the location of areas not suitable for development, which are to be avoided during construction and operation, where relevant;	Section 5.3
2.7.7	additional environmental impacts expected from the proposed development;	Section 5.1
2.7.8	any direct, indirect and cumulative impacts of the proposed development on site;	Section 5.1
2.7.9	the degree to which impacts and risks can be mitigated;	Section 5
2.7.10	the degree to which the impacts and risks can be reversed;	Section 5
2.7.11	the degree to which the impacts and risks can cause loss of irreplaceable resources;	
2.7.12	a suitable construction and operational buffer for the aquatic ecosystem, using the accepted methodologies;	Section Error! Reference source not found.

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2.7.13	proposed impact management actions and impact management outcomes for inclusion in the Environmental Management Programme (EMPr);	Section 5.3
2.7.14	a motivation must be provided if there were development footprints identified as per paragraph 2.4 above that were identified as having a “low” aquatic biodiversity sensitivity and that were not considered appropriate;	N/A
2.7.15	a substantiated statement, based on the findings of the specialist assessment, regarding the acceptability or not of the proposed development and if the proposed development should receive approval or not; and	Section 6
2.7.16	any conditions to which this statement is subjected.	N/A

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ABBREVIATIONS

CBA	Critical Biodiversity Areas
DWS	Department of Water & Sanitation
ECBCP	Eastern Cape Biodiversity Conservation Plan
EAP	Environmental Assessment Practitioner
EI&S	Ecological Importance and Sensitivity
ESA	Ecological Support Area
I&APs	Interested and Affected Parties
IHI	Index for Habitat Integrity
IUA	Integrated Unit of Analysis
NFEPA	National Freshwater Ecosystem Priority Areas
NEMA	National Environmental Management Act
PA	Protected Areas
PES	Present Ecological State
SANBI	South African National Biodiversity Institute
SCC	Species of Conservation Concern
SQR	Sub-Quaternary Reach
WMA	Water Management Area
WULA	Water Use Licence Application

1. INTRODUCTION

Abantu Environmental Services (Pty) Ltd was appointed to undertake an aquatic biodiversity assessment as part of an Environmental Authorisation (EA) process for the proposed construction of a conference centre in Dutywa, within the Mbhashe Local Municipality (**Figure 1**). This report presents the aquatic biodiversity impact assessment report for the development.

1.1 PROJECT DESCRIPTION

The proposed development comprises of the following:

- Proposed conference centre
- Proposed Sewer Pipe - 160mm Class 34 UPVC and Galvanised Steel Sewer Pipe
- Proposed Water Pipe - main - 8 m servitude
- Proposed Concrete Stormwater Pipes- 450mm – 600mm Ø 100D
- Proposed Sewer Manholes
- Proposed Sw Catchpit/Manhole
- **Proposed concrete/paved/tar access road and a crossing structure (bridge/culvert)**
 - 5960mm (5.96km) length
 - width of 3700mm x 2 lanes (7.4m + 1.65m = 9.5m)
 - shoulders including sidewalks;
 - sidewalks = 1200mm
 - stormwater drains and should of 465mm
 - **Layers works**
 - 60mm thick 35mpa interlocking brick
 - paving over 20mm compacted sand
 - 150mm g5 material stabilised with 3%
 - cement compacted to 98% mod aashto
 - 150mm g7 material compacted to
 - 95% mo150mm insitu compacted to
 - 93% mod aashtod aashto

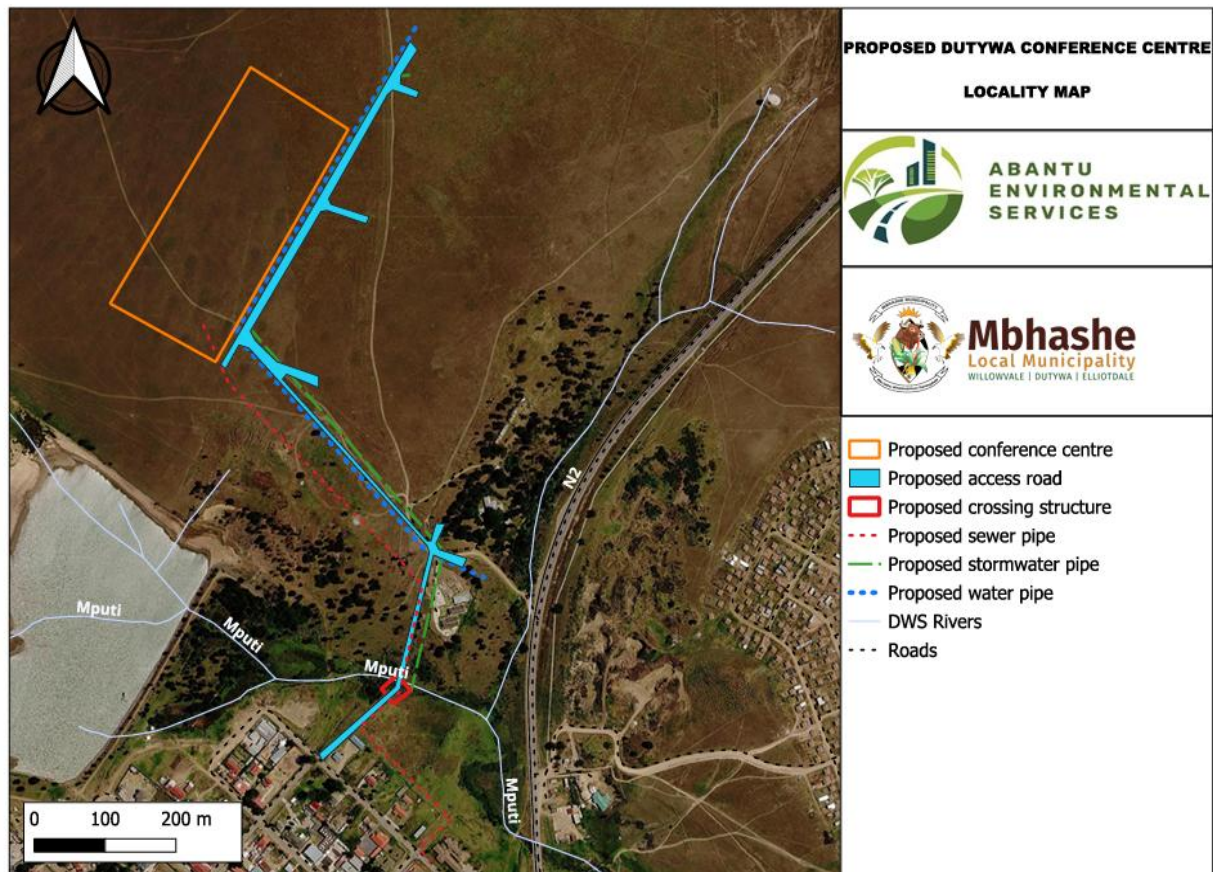


Figure 1. Locality map

1.2 LEGISLATIVE REQUIREMENTS

According to the protocols specified in GN 320 (Protocol for the specialist assessment and minimum report content requirements for environmental impacts on aquatic biodiversity) of the National Environmental Management Act (NEMA; Act No. 107 of 1998), assessment and reporting requirements for aquatic biodiversity are associated with a level of environmental sensitivity identified by the national web-based environmental screening tool (screening tool). An applicant intending to undertake an activity identified in the scope of this protocol on a site identified by the screening tool as being of:

- **Very High** sensitivity for aquatic biodiversity, must submit an Aquatic Biodiversity Specialist Assessment; or
- **Low** sensitivity for aquatic biodiversity, must submit an Aquatic Biodiversity Compliance Statement.

The screening tool classified the aquatic biodiversity theme for the study area as **Very High** sensitivity, with mapped sensitivity features including aquatic ESA 1 and Wetlands_Seep.

According to the protocol, prior to commencing with a specialist assessment a site sensitivity verification must be undertaken to confirm the sensitivity of the site as indicated by the screening tool:

- Where the information gathered from the site sensitivity verification differs from the screening tool designation of **Very High** aquatic biodiversity sensitivity, and it is found to be of a **Low** sensitivity, an Aquatic Biodiversity Compliance Statement must be submitted.

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Similarly, where the information gathered from the site sensitivity verification differs from the screening tool designation of **Low** aquatic biodiversity sensitivity, and it is found to be of a **Very High** sensitivity, an Aquatic Biodiversity Specialist Assessment must be submitted.

The following legislation is directly relevant when assessing the aquatic environment relating to the proposed project.

Table 4. Applicable legislation

Legislation / guideline	Application to the project
National Environmental Management Act (NEMA) (Act 107 of 1998), as amended	Framework environmental management principles and duty of care informing avoidance/mitigation and sustainable development decision-making.
EIA Regulations, 2014 (as amended) and Listing Notices (as amended)	Project triggers listed activities requiring Environmental Authorisation; prescribes public participation and specialist report requirements.
National Water Act (Act 36 of 1998)	Determines whether the works trigger Section 21 water uses (commonly 21(c) and/or 21(i) for in channel/riparian works; and potentially discharge related uses where applicable); includes pollution prevention duty (Section 19).
General Authorisations / WULA requirements (where applicable)	Confirms whether water uses may be authorised under GA or require WULA and informs licence/GA conditions relevant to aquatic protection.
GN 320 (2020): Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Aquatic Biodiversity	Governs aquatic biodiversity specialist assessment process and minimum report content, including Screening Tool sensitivity-driven outputs and site sensitivity verification.
NEM: Biodiversity Act (NEMBA) (Act 10 of 2004)	Provides for protection of listed threatened/protected species and invasive species management relevant to aquatic/riparian habitats.
Eastern Cape Biodiversity Conservation Plan (ECBCP)	Spatial biodiversity planning layer identifying CBAs/ESAs guiding avoidance and mitigation in sensitive areas.
National Freshwater Ecosystem Priority Areas (NFEPA)	Freshwater ecosystem priority mapping to contextualise ecological importance and support sensitivity/risk motivation.

1.3 ASSUMPTIONS AND LIMITATIONS

The findings and recommendations of this report may be susceptible to the following uncertainties and limitation:

- The report is based on currently available information and, as a result, limited by the information provided by the client.
- Aquatic ecosystems vary both temporally and spatially. Once-off surveys such as this are therefore likely to miss certain ecological information due to seasonality, thus limiting accuracy and confidence.
- The field assessment was undertaken during July 2026 and therefore represents the condition of the watercourse at the time of the survey only.
- No detailed assessment of aquatic biota, including fish, aquatic macroinvertebrates and aquatic vegetation, was undertaken.
- The scope of work did not include in situ or laboratory water quality sampling.

2. METHODOLOGY

The aim of this assessment is to identify areas of aquatic importance and to evaluate these in terms of their conservation importance. In line with the assessment and reporting requirements set out in the protocol, this report includes two main study components: a desktop literature review, supplemented by a field survey within the proposed development footprint and extended areas of influence.

2.1 DESKTOP ASSESSMENT

The aim of the desktop literature review component was to collate and review the extensive available ecological information related to important biodiversity and conservation features and key ecological processes and function.

The following sources were consulted for the desktop literature review.

Table 5. Utilised data and associated source

Data	Source
Google Earth Pro	Google Earth Pro
National Screening Tool	DFFE
DWS Ecoregions	DWA (2005)
Eastern Cape Biodiversity Conservation Plan (ECBCP)	South African National Biodiversity Institute (SANBI), obtained from BGIS
Desktop Present Ecological State, Ecological Importance and Ecological Sensitivity per Sub Quaternary Reaches for Secondary Catchments in South Africa	Resource Quality Information Service (RQIS) DWS (2014)
National Freshwater Ecosystem Priority Areas (NFEPA)	CSIR (2011), obtained from BGIS
National Vegetation Map	SANBI obtained from BGIS
National Wetland Map 5	CSIR and SANBI

2.2 ASSESSMENT METHODS/TOOLS

The site assessment was conducted on 28 July 2026 to verify the aquatic biodiversity sensitivity identified by the national web-based environmental screening tool and to identify any discrepancies between the mapped sensitivity and the current environmental conditions observed in the field. The following techniques and methodologies were utilised to undertake the assessments:

2.2.1 PRESENT ECOLOGICAL STATE

Drainage lines and rivers are natural channels in which water flows permanently or intermittently following rainfall. These are assessed using the Index of Habitat Integrity (IHI; Kleynhans, 1996) which measures the impact of human disturbance on riparian and instream habitats. The IHI is a rapid assessment of the severity of impacts affecting habitat integrity within a defined segment of a watercourse. The method can be applied to both perennial and non-perennial watercourses. The instream impacts considered were

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water abstraction; flow modification; bed modification; channel modification; physio-chemical modification; inundation; alien macrophytes; and rubbish dumping. The riparian impacts assessed were vegetation removal; exotic vegetation; bank erosion; channel modification; water abstraction; inundation; flow modification; physico-chemistry. Each of the impacts was given a score based on their degree of modification, along with a confidence rating based on the level of confidence in the score.

Table 6. Descriptive classes for assessment of habitat modifications (Kleynhans, 1996)

Impact Class	Description	Score
None	No discernible impact or the modification is located in such a way that it has no impact on habitat quality, diversity, size and variability.	0
Small	The modification is limited to very few localities and the impact on habitat quality, diversity, size and variability are also very small.	1-5
Moderate	The modifications are present at a small number of localities and the impact on habitat quality, diversity, size and variability are also limited.	6-10
Large	The modification is generally present with a clearly detrimental impact on habitat quality, diversity, size and variability. Large areas are, however, not influenced.	11-15
Serious	The modification is frequently present and the habitat quality, diversity, size and variability in almost the whole of the defined area are affected. Only small areas are not influenced.	16-20
Critical	The modification is present overall with a high intensity. The habitat quality, diversity, size and variability in almost the whole of the defined section are influenced detrimentally.	21-25

Each of the allocated scores are then moderated by a weighting system (Table 7), which is based on the relative threat of the impact to the habitat integrity of the riverine system. The total score for each impact is equal to the assigned score multiplied by the weight of that impact. The estimated impacts (assigned score / maximum score [25] X allocated weighting) of all criteria are then summed together, expressed as a percentage and then subtracted from 100 to determine the Present Ecological State score (or Ecological Category) for the instream and riparian components, respectively.

Table 7. Criteria and weights used for the assessment of habitat integrity (Kleynhans, 1996)

Instream Criteria	Weight	Riparian Zone Criteria	Weight
Water abstraction	14	Indigenous vegetation removal	13
Flow modification	13	Exotic vegetation encroachment	12
Bed modification	13	Bank erosion	14
Channel modification	13	Channel modification	12
Water quality modification	14	Water abstraction	13
Inundation	10	Inundation	11
Alien/Exotic macrophytes	9	Flow modification	12
Alien/Exotic aquatic fauna	8	Water quality	13
Solid waste disposal	6		

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Total	100	Total	100
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An IHI class is then determined based on the resulting score. These results provide an indication of the site-specific PES which can be used as a baseline from which to monitor impacts in the future.

Table 8. Index of habitat integrity classes and descriptions

Integrity Class	Description	Score
A	Unmodified, natural.	90-100
B	Largely natural with few modifications. The flow regime has been only slightly modified and pollution is limited to sediment. A small change in natural habitats may have taken place. However, the ecosystem functions are essentially unchanged.	80-89
C	Moderately modified. Loss and change of natural habitat and biota have occurred, but the basic ecosystem functions are still predominantly unchanged.	60-79
D	Largely modified. A large loss of natural habitat, biota and basic ecosystem functions has occurred.	40-59
E	Seriously modified. The loss of natural habitat, biota and basic ecosystem functions is extensive.	20-39
F	Critically / Extremely modified. Modifications have reached a critical level and the system has been modified completely with an almost complete loss of natural habitat and biota. In the worst instances the basic ecosystem functions have been destroyed and the changes are irreversible.	0-19

2.2.2 ECOLOGICAL IMPORTANCE AND SENSITIVITY

The ecological importance of a river is an expression of its importance to the maintenance of biological diversity and ecological functioning on local and wider scales. Ecological sensitivity (or fragility) refers to the system's ability to resist disturbance and its capability to recover from disturbance once it has occurred (resilience). Both abiotic and biotic components of the system are taken into consideration in the assessment of ecological importance and sensitivity (Kleynhans & Louw, 2007).

The Ecological Importance and Sensitivity (EIS) assessment considers several biotic and habitat determinants surmised to indicate either importance or sensitivity. The determinants are rated according to a four-point scale. The median of the resultant score is calculated to derive the EIS category.

Table 9. Scale used to assess determinants presumed to indicate Importance or Sensitivity

Scale	Definition
1	One species/taxon judged as rare or endangered at a local scale.
2	More than one species/taxon judged to be rare or endangered on a local scale.
3	One or more species/taxon judged to be rare or endangered on a Provincial/regional scale.
4	One or more species/taxon judged as rare or endangered on a National scale (SA Red Data Books)

Table 10. Ecological Importance and Sensitivity Categories

EISC	General Description	Median
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Very High	Quaternaries/delineations unique on a national and international level based on unique biodiversity. These rivers are usually very sensitive to modifications.	>3-4
High	Quaternaries/delineations considered unique on national scale based on biodiversity. These rivers may be sensitive to modifications but have capacity for use.	>2-≤3
Moderate	Quaternaries/delineations considered unique on a provincial or local scale due to biodiversity. These rivers are not usually very sensitive to modifications.	>1-≤2
Low/marginal	Quaternaries/delineations not unique on any scale. These rivers are generally not very sensitive to flow modifications and usually have substantial capacity for use.	≤1

2.3 SAMPLING SITES

A total of four assessment sites were selected along the Mputi River to characterise upstream, crossing and downstream conditions. The site locations, coordinates and brief descriptions are provided in **Table 11**.

Table 11. Sampling points

River	Site	COORDINATES	Description
Mputi River	MR 1	32° 05'38.82"S; 28°17'35.87"E	Site located in the upper reach of the assessed Mputi River, upstream of the proposed crossing.
	MR 2	32° 05'29.64"S; 28°18'20.69"E	Site located at the proposed crossing on the Mputi River.
	MR 3	32° 05'37.53"S; 28°18'29.01"E	Site located downstream of the proposed crossing at the N2 road bridge.
	MR 4	32° 06'10.91"S; 28°18'50.46"E	Site located further downstream at the confluence of the Mputi River and the Ngxakaxa River.

2.4 IMPACT ASSESSMENT METHODOLOGY

The criterion for determining impact significance has been defined in accordance with the criteria drawn from Appendix 3 of the Environmental Impact Assessment Regulations, 2014. The levels of detail described in the EIA regulations were fine-tuned by assigning specific values to each impact identified. The impact ratings are informed by fieldwork, desktop analysis and expertise of the specialist. The significance of potential impacts that may result from the proposed development will be determined in order to assist the competent authority in making a decision.

Status of impact	
Indication whether the impact is adverse (negative) or beneficial (positive).	+ ve (positive – a 'benefit')
	- ve (negative – a 'cost')

Rating	Definition of Rating	Score
A. Spatial extent – the area in which the impact will be experienced		

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None		0
Local	Confined to project or study area or part thereof (e.g. site)	1
Regional	The region, which may be defined in various ways, e.g. cadastral, catchment, topographic	2
(Inter) national	Nationally or beyond	3
B. Intensity– the magnitude or size of the impact		
None		0
Low	Natural and/or social functions and processes continue albeit in a slightly modified way	1
Medium	Natural and/or social functions and processes continue albeit in a modified way	2
High	Natural and/or social functions or processes are severely altered	3
C. Duration– the time frame for which the impact will be experienced		
None		0
Short-term	Up to 2 years	1
Medium-term	2 to 20 years	2
Long-term	More than 20 years	3

Combined Score (A+B+C)	
Consequence Rating	
Not significant	0 – 2
Very low	3 – 4
Low	5
Medium	6
High	7
Very high	8 – 9

Probability– the likelihood of the impact occurring		
Improbable	< 40% chance of occurring	0
Possible	40% - 70% chance of occurring	1
Probable	> 70% - 90% chance of occurring	2
Definite	> 90% chance of occurring	3

Significance Rating	Possible Impact Combinations	
	Consequence	Probability
Insignificant	Very Low	Improbable
	Very Low	Possible
Very Low	Very Low	Probable
	Very Low	Definite
Low	Low	Improbable
	Low	Possible
	Low	Probable
	Low	Definite
Medium	Medium	Improbable
	Medium	Possible
	Medium	Probable
	Medium	Definite
High	High	Improbable
	High	Possible
	High	Probable
	High	Definite

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Significance Rating	Possible Impact Combinations	
	Consequence	Probability
	Very High	Improbable
	Very High	Possible
Very High	Very High	Probable
	Very High	Definite

Confidence of assessment	Rating
The degree of confidence in predictions based on available information, Specialists' judgment and/or specialist knowledge.	Low
	Medium
	High

3. AQUATIC BIODIVERSITY DESKTOP ASSESSMENT

3.1 TOPOGRAPHY

The area is characterised by undulating to gently rolling inland terrain (see **Figure 2**), with the broader municipality displaying fragmented topography of open grasslands, valleys and river corridors. It is generally less rugged than the coastal parts although it is still associated with incised drainage lines, eroded slopes and uneven landforms in places (MLM IDP, 2023).

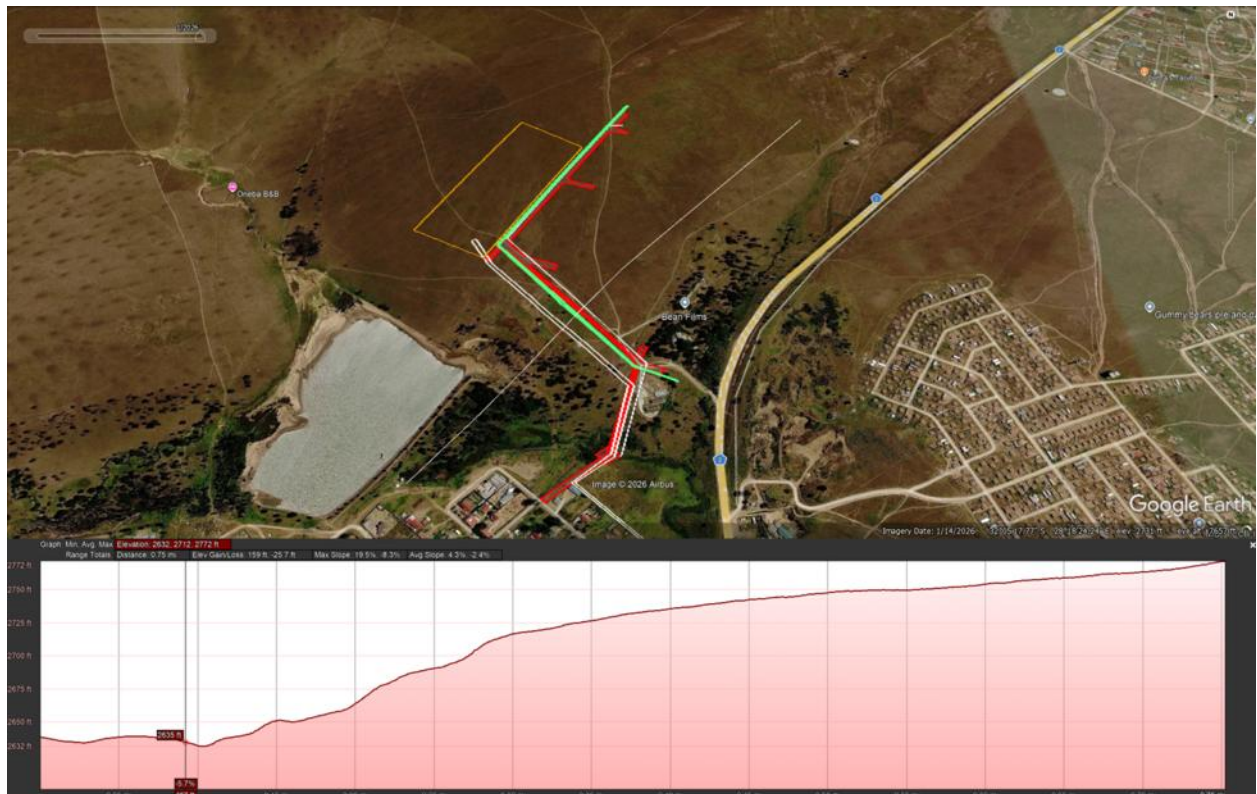


Figure 2. Topographic profile of the development site.

3.2 CLIMATE

Dutywa is situated at an altitude of approximately 388m above sea level and receives approximately 530 – 540 mm of rainfall annually, with March being the wettest month. The area experiences a warm temperate climate with summer rainfall, characterized by warm summers and cool, dry winters. The warmest conditions are generally experienced during February, with an average midday temperature of about 25°C, while the coldest condition occur during July, when average temperatures drop to around 18.3°C during the day and 5.1°C at night.

3.3 GEOLOGY AND SOILS

The area is dominated by sedimentary rocks of the Karoo Supergroup, mainly of Late Permian to Early Triassic age, together with younger Jurassic dolerite intrusions. The area is predominantly underlain by

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the Beaufort Group, particularly the Adelaide Subgroup. The broader Dutywa area is also associated with the Katberg Formation of the Tarkastad Subgroup, which generally comprises of red and greenish-grey mudstone with subordinate fine-to-medium grained sandstone. In some parts of the wider Dutywa area, younger alluvial deposits are also associated with drainage lines and river valleys.

The soils in the Dutywa area are generally derived from the weathering of the underlying mudstone, sandstone and dolerite, and are typically shallow to moderately deep, depending on slope and proximity to drainage features. These soils may be prone to erosion where vegetation has been disturbed or where they occur on steeper slopes, while deeper alluvial soils may occur along valley bottoms.

3.4 VEGETATION

According to South African National Biodiversity Institute (SANBI) vegetation map (VegMap,2018) the proposed site falls within the Mthatha Moist Grassland, under the Grassland Biome. This vegetation type is typically made up of species-poor, sour, wiry grassland with *Eragrostis plana* and *Sporobolus africanus*. The grassland is in good condition, but likely to be dominated by *Themeda triandra* (Mucina et al., 2006). The Mthatha Moist Grassland is classified as Endangered according to The Revised National List of Threatened Terrestrial Ecosystems –2022.

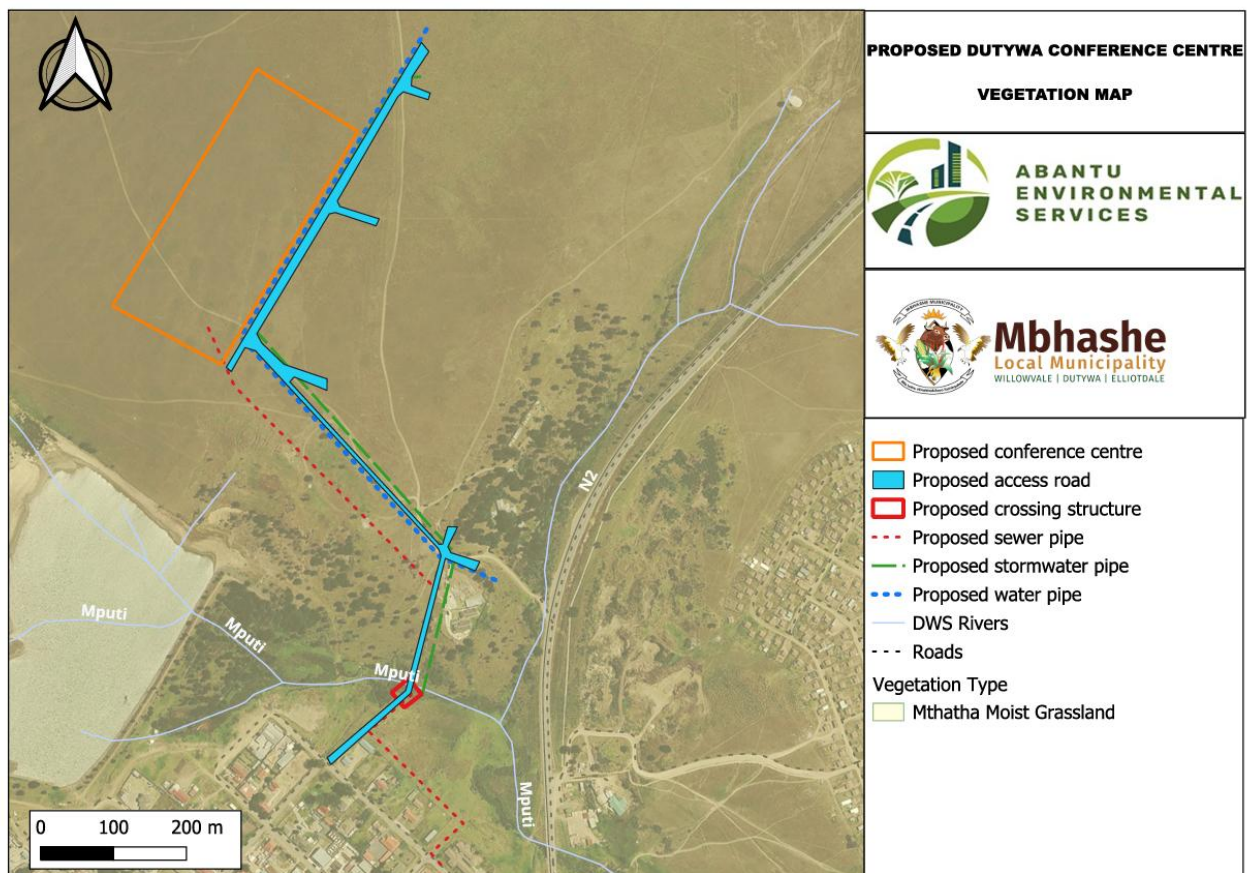


Figure 3. Vegetation Map

3.5 BIODIVERSITY CONSERVATION PLANS- EASTERN CAPE BIODIVERSITY CONSERVATION PLAN, 2019

The Eastern Cape Biodiversity Conservation Plan (ECBCP) (2019) replaces the ECBCP (2007) in its entirety and provides a map of important biodiversity areas, outside of the Protected Areas network, which must be used to inform land use and resource-use planning and decision making. The objectives of the ECBCP (2019) are to:

- 1) Identify the minimum spatial requirements needed to maintain a living landscape that continues to support all aspects of biodiversity and retain/maintain essential ecological infrastructure. This is achieved through the selection of areas, based on achieving targets, which represent important biodiversity pattern AND ecological processes;
- 2) Serve as the primary source of biodiversity information for land use planning and decision making; and
- 3) Inform conservation and restoration action in important biodiversity areas.

The aim of the ECBCP (2019) was to map biodiversity priority areas through a systematic conservation planning process. The main outputs of the ECBCP include Protected Areas (PA), Critical Biodiversity Areas (CBA), Ecological Support Areas (ESA), Other Natural Areas (ONA) and No Natural Habitat Remaining (NNR) for both terrestrial and aquatic ecosystems. The ECBCP (2019) has been adopted by DEDEAT as a systematic biodiversity plan for the Eastern Cape Province.

According to the ECBCP (2019), the proposed development site is located within aquatic ESA 1 (see **Figure 4**).

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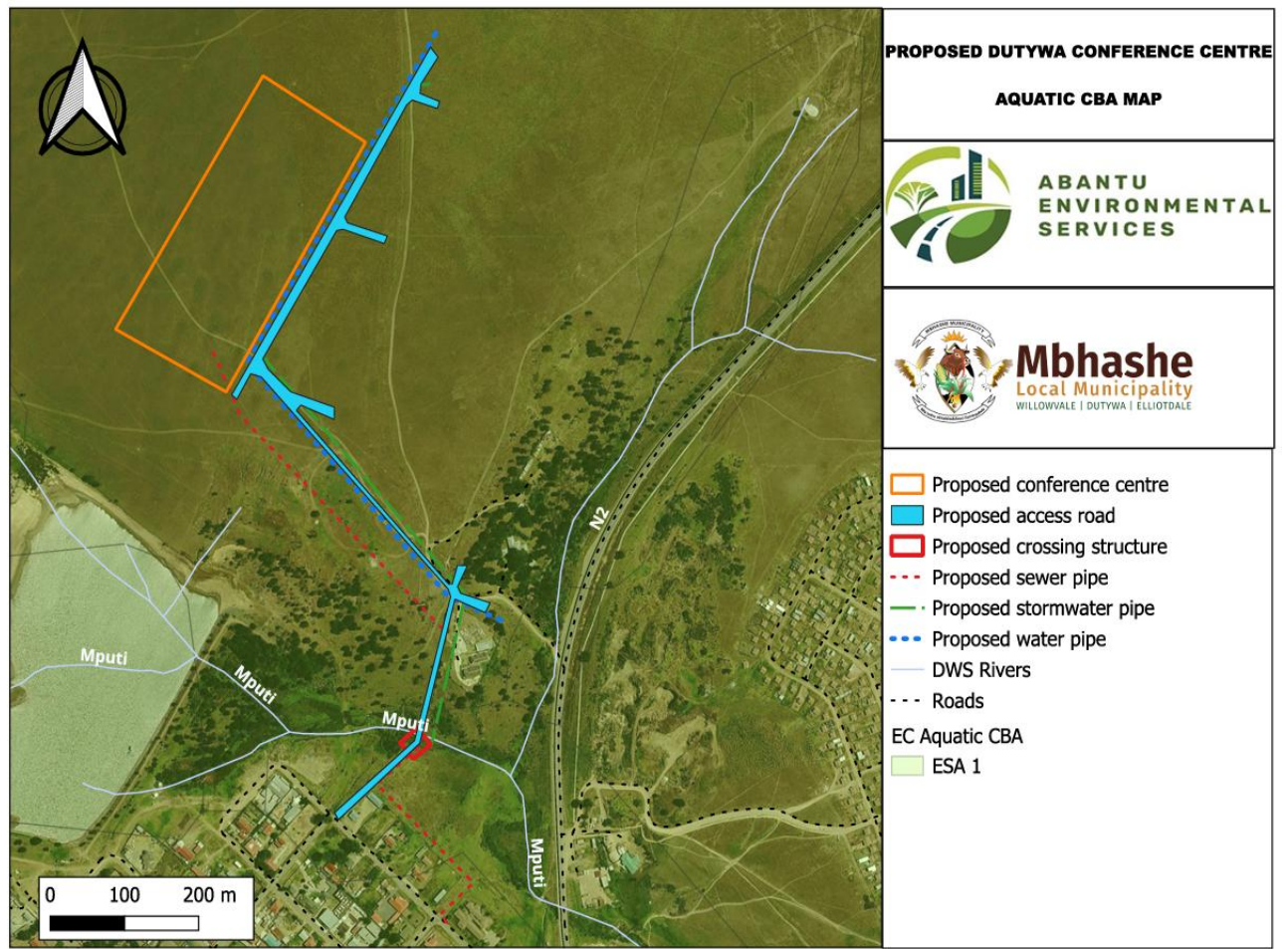


Figure 4. Aquatic Map

3.6 FRESHWATER ECOLOGY

The project area falls within the Mzimvubu- Tsitsikamma Water Management Area, within T90A quaternary catchment. The T90A catchment area falls within the South Eastern Uplands Level 1 Ecoregion (16.06) Level 2 Ecoregion. The Mputi River forms the primary watercourse within the immediate vicinity of the proposed development. The river flows generally south-west from the project area before joining the Ngxakaxa River, after which the combined flow drains downstream into the Nqabara River (see **Figure 5**). The Mputi River is not represented as a separate Sub-Quaternary Reach (SQR) within the DWS desktop PES/EI/ES dataset; therefore, no desktop PES/EIS information is available specifically for the Mputi River.

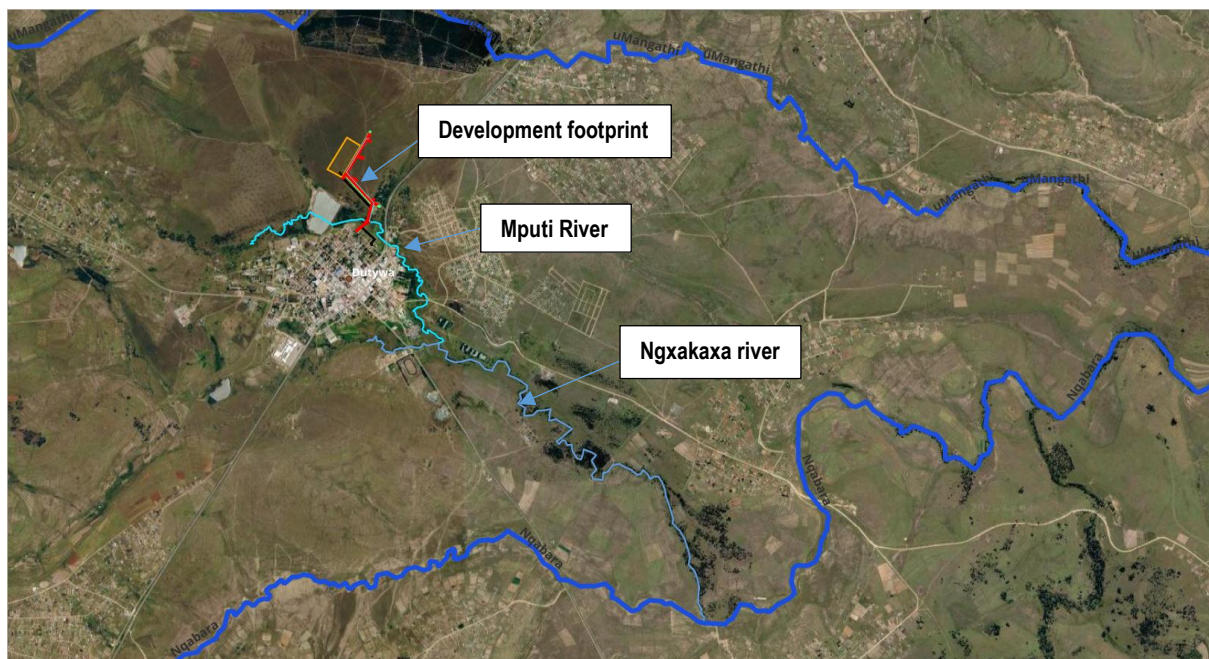


Figure 5. Hydrological connectivity of the Mputi River within the study area, showing its confluence with the Ngxakaxa River and downstream drainage to the Nqabara River.

3.7 NATIONAL FRESHWATER ECOSYSTEM PRIORITY AREAS (NFEPA)

The Atlas of Freshwater Ecosystem Priority Areas in South Africa (Nel et al, 2011a) (The Atlas) which represents the culmination of the National Freshwater Ecosystem Priority Areas project (NFEPA), a partnership between SANBI, CSIR, WRC, DEA, DWA, WWF, SAIAB and SANParks, provides a series of maps detailing strategic spatial priorities for conserving South Africa's freshwater ecosystems and supporting sustainable use of water resources.

Freshwater Ecosystem Priority Areas (FEPA's) were identified through a systematic biodiversity planning approach that incorporated a range of biodiversity aspects such as ecoregion, current condition of habitat, presence of threatened vegetation, fish, frogs and birds, and importance in terms of maintaining

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downstream habitat. The Atlas incorporates the National Wetland Inventory (NWI Wetlands) (SANBI, 2011) to provide information on the distribution and extent of wetland areas. River, wetland and estuarine FEPAs should be regarded as significant water resources and should be regarded as ecologically important and as generally sensitive to changes in water quality and quantity, owing to their role in protecting freshwater ecosystems and supporting sustainable use of water resources.

According to the NFEPA datasets the development does not overlap any FEPA rivers. The NFEPA dataset identifies a mapped wetland feature within 500 m of the proposed development footprint (see **Figure 6**). Field verification confirmed that the mapped aquatic feature within this area comprises an artificial dam/impoundment rather than a natural wetland system.

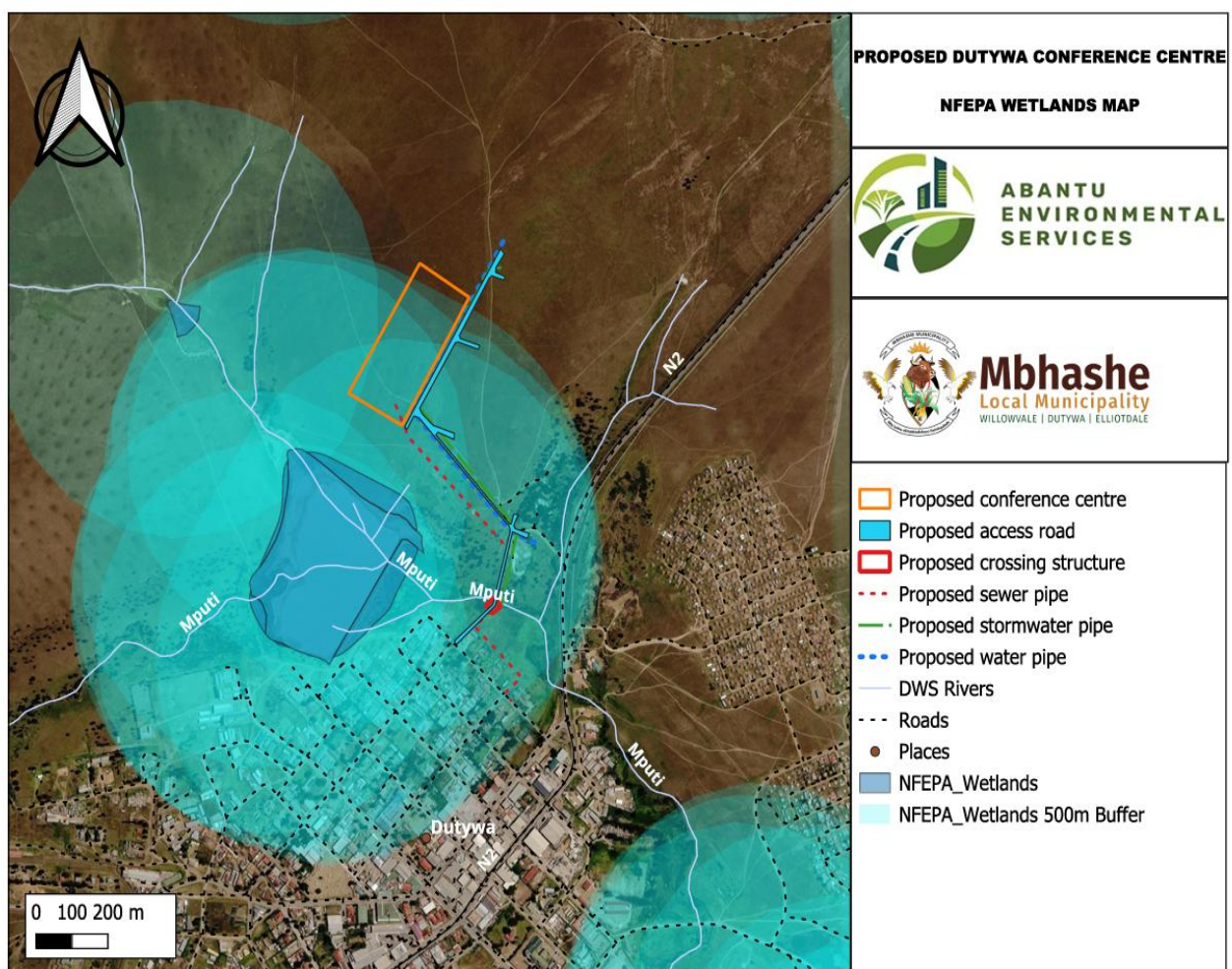


Figure 6. NFEPA wetland map

3.8 NATIONAL WETLAND MAP 5 (NWM5)

The National Wetland Map version 5 (NWM5) shows the distribution of inland wetland ecosystem types across South Africa and includes estuaries and the extent of some rivers. A confidence map was compiled to identify areas where wetland extent and hydrogeomorphic (HGM) units (which contributed to defining the inland wetland ecosystem types together with the regional setting) attained at a higher level of certainty compared to other areas.

Higher levels of certainty are associated [code 5 in field Confidence_nr] with areas that have been visited in field by a wetland specialist(s) over multiple seasons and cycles of the wetland hydroperiod and are therefore more accurately represented in the dataset. Codes 4 to 1 indicate lower levels of confidence that the extent and HGM unit are represented well.

The National Wetland Map version 5 (NWM5) shows no inland wetland ecosystems.

3.9 STRATEGIC WATER SOURCE AREA

Strategic Water Source Areas (SWSAs) are now defined as areas of land that either: (a) supply a disproportionate (i.e. relatively large) quantity of mean annual surface water runoff in relation to their size and so are considered nationally important; or (b) have high groundwater recharge and where the groundwater forms a nationally important resource; or (c) areas that meet both criteria (a) and (b). They include transboundary Water Source Areas that extend into Lesotho and Swaziland. All surface water SWSAs are located in high rainfall areas where baseflow is at least 11 25 mm/a, which is evidence of a strong link between groundwater and surface water in the SWSAs. The aquifers sustain baseflow, contribute to runoff and, especially, contribute to dry season flows. Sustained river flows are important as they support people and communities who depend directly on rivers for their water, especially during the dry season and droughts.

The proposed development footprint does not fall within any SWSA.

3.10 RESOURCE QUALITY OBJECTIVES

Resource Quality Objectives (RQOs) are a set of narrative and/or numerical management objectives defined for any particular resource, once the class is determined. RQOs encompass four components of the resource namely: Water quantity; Water quality; Habitat integrity; and Biotic characteristics. RQOs are important management objectives against which resource monitoring will be assessed and form important sustainability indicators for water resource management (DWS, 2022).

The Department of Water and Sanitation (DWS) has undertaken the determination of Water Resource Classes, the Reserve and Resource Quality Objectives for the Keiskamma and Fish-to-Tsitsikamma

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catchments. The DWS Water Resource Classification System delineates the project area within Integrated Unit of Analysis IUA_T04: Pondoland coastal, which includes the T90A quaternary catchment. The Nqabarha River is also represented within this IUA. The Mputi River which forms the primary watercourse associated with the proposed development, drains into the Ngxakaxa River and subsequently into the Nqabarha River. No site-specific RQOs were identified for the Mputi River during the review of available DWS documentation. The RQOs established for the broader IUA and downstream Nqabarha River system therefore provide relevant management context for the assessment but should not be interpreted as site specific numerical RQOs for the assessed Mputi River reach.

4. SITE SPECIFIC FINDINGS

The sections below present the results of the aquatic biodiversity assessment undertaken on 27 June 2026, during the winter season. Low-flow conditions were observed within the Mputi River at the time of the assessment. The field assessment confirmed the presence and location of the Mputi River, which is mapped as a non-perennial watercourse in the DWS 1:50 000 NGI river dataset. Field verification also confirmed the presence of an artificial dam/impoundment, the dam was considered in the assessment due to its proximity to the proposed development and its potential susceptibility to indirect construction-related impacts.

4.1 FLOW CONDITIONS AND HABITAT OBSERVATIONS

Four assessment points were selected along the Mputi River to characterise flow conditions, channel characteristics and associated aquatic and riparian habitat features. The locations of the assessed points are presented in **Table 12**. The assessed Mputi River reach occurs within a Lower Foothill geomorphic zone and was generally characterised by a narrow and shallow channel. Surface water was present at the assessed locations; however, overall low-flow conditions prevailed, with slow to moderate flow observed during the assessment.

Table 12. Flow conditions at the assessed river reach

River		Description
MR 1- Upstream		
		This upstream site exhibited low-flow conditions, with shallow surface water confined to a narrow channel. An informal crossing was observed, with associated localised disturbance of the channel bed and banks.

MR 2- At the proposed crossing



This site exhibited shallow, slow-flowing surface water within a narrow channel. Relatively dense marginal and riparian vegetation occurred along the channel margins, with localised bank erosion and sediment deposition evident.

MR 3- N2 bridge



This site is located at the N2 bridge crossing, where slow-flowing surface water was observed. The existing road crossing and damaged concrete structure have resulted in localised modification of the channel bed and banks, with associated alteration of channel morphology.

MR 4- At the confluence

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This site represents the confluence of the Mputi River and Ngxakaxa River. Slow-moving surface water was observed at the time of the assessment. Anthropogenic disturbance was evident, particularly through the accumulation of solid waste within the channel and riparian vegetation, together with localised bank disturbance.

4.2 HABITAT ASSESSMENT

Habitat quality and availability plays a critical role in the occurrence of aquatic biota. For this reason, habitat evaluation is conducted simultaneously with biological evaluations in order to facilitate the interpretation of results (Ollis et al., 2006). The quality of the instream and riparian habitat influences the structure and function of the aquatic community in a stream; therefore, assessment of the habitat is critical to any assessment of ecological integrity.

In the current study, the instream and riparian habitat assessment was based on the Index of Habitat Integrity (IHI).

4.2.1 INDEX OF HABITAT INTEGRITY

The IHI is a tool developed to assess river habitat integrity and forms part of the River EcoStatus Monitoring Program (REMP) (Kleynhans, 2007a). The evaluation of Habitat Integrity provides a measure of the degree to which a river has been modified from its natural state, in other words, an indication of the present ecological state (PES) of the watercourse. A desktop level and reach-based Habitat Integrity (IHI) was applied and observations made during the field survey were used to supplement the data used within the index. Results are presented in **Table 13** and **Table 14**.

Table 13. Instream habitat integrity

Instream impact	Score (0-25)	Weight	Weighted impact
Water abstraction	10	14	5.6
Flow modification	8	13	4.16
Bed modification	7	13	3.64
Channel modification	10	13	5.2
Physio-chemical modification	7	14	3.92
Inundation	0	10	0.00
Alien/Exotic macrophytes	0	9	0.00
Alien/Exotic aquatic fauna	0	8	0.00
Rubbish dumping	13	6	3.12
IHI Instream score	74		
Integrity class:	C		

Table 14. Riparian habitat integrity

Riparian impact	Score (0-25)	Weight	Weighted impact
Indigenous vegetation removal	5	13	2.60
Exotic vegetation encroachment	2	12	0.96
Bank erosion	10	14	5.60
Channel modification	7	12	3.36
Water abstraction	3	13	1.56
Inundation	0	11	0.00
Flow modification	4	12	1.92
Physio-chemical modification	3	13	1.56

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IHI Riparian score	81
Integrity class:	B

The Index of Habitat Integrity (IHI) assessment for the Mputi River indicates that the instream component falls within Integrity Class C (Moderately Modified), while the riparian component falls within Integrity Class B (Largely Natural with Few Modifications). The instream habitat score of approximately 74% indicates that the channel remains functional and retains several natural characteristics, but has been moderately modified by water abstraction, flow-related alteration, localised bed and channel modification, and anthropogenic pressures, including solid waste accumulation. The riparian habitat score of approximately 81% indicates that the riparian zone remains largely natural, with modifications primarily associated with localised vegetation removal, bank erosion, channel disturbance and other anthropogenic activities.

4.2.2 ECOLOGICAL IMPORTANCE AND SENSITIVITY

The Ecological Importance and Sensitivity (EIS) of the assessed Mputi River reach was evaluated using the Kleynhans (1999) approach, which considers a range of biotic and aquatic habitat determinants. The overall EIS category was derived from the median of the determinant scores. The results of the assessment are presented in **Table 15**. The ecological importance and sensitivity category of the Mputi River reach was determined as being Moderate.

Table 15. Results of the EIS assessment for Mputi River

Determinants	Score (0-4)
Rare and endangered biota	0
Unique biota	0
Intolerant biota	1
Species/taxon richness	1
Diversity of aquatic habitat types or features	2
Refuge value of habitat types	2
Sensitivity of habitat to flow changes	2
Sensitivity to flow related water quality changes	1
Migration route/corridor for instream and riparian biota	2
National parks, Wilderness areas, Nature reserves, Natural Heritage sites, and natural area	2
EIS score	1.5
EIS category	Moderate

5. IMPACT ASSESSMENT

5.1 IDENTIFIED POTENTIAL IMPACTS

The following potential aquatic ecosystem impacts have been identified for the proposed Dutywa conference centre (Table 16).

Table 16. Identified impacts

Impact	Description
Construction Phase	
Non-compliance with environmental and water legislation	Development commencement without the required environmental approvals and water use authorisation may result in legal non-compliance, project delays, and possible enforcement action against the municipality and contractors.
Loss/disturbance of riparian vegetation and associated habitat	Clearing and disturbance of riparian vegetation during construction may result in the localised loss of vegetation cover and associated habitat, with potential effects on bank stability and habitat structure.
Disturbance and modification of the riverbed and banks	Earthworks, excavation, machinery movement and construction of the proposed crossing and associated infrastructure may physically disturb the riverbed and banks and result in localised alteration of the natural channel form and habitat characteristics.
Bank erosion and sedimentation	Vegetation clearing, exposed soils, earthworks and concentrated runoff may increase erosion of the riverbanks and surrounding disturbed areas, resulting in increased sediment input and deposition within the Mputi River and potentially within the nearby artificial dam/impoundment through sediment-laden runoff.
Flow modification	Construction of the proposed crossing may alter the natural movement and distribution of surface flow within the Mputi River through temporary obstruction, diversion or concentration of flow. If not appropriately designed and constructed, the crossing may also cause localised scour or sediment deposition and affect longitudinal ecological connectivity within the river.
Deterioration of surface water quality	Construction activities may introduce sediment, hydrocarbons, chemicals, litter or other pollutants into the river through accidental spills, runoff or inappropriate waste handling, potentially affecting aquatic habitat.
Cumulative impact on aquatic biodiversity	The proposed development may contribute cumulatively to existing pressures on the Mputi River, including road crossings, stormwater inputs, bank erosion and solid-waste accumulation. Additional habitat disturbance, runoff, sedimentation and pollution associated with the development may therefore contribute to progressive degradation of instream and riparian habitat.
Operation Phase	
Increased stormwater runoff	Permanent hard surfaces may increase the volume and concentration of stormwater runoff during rainfall events, increasing erosion and the potential for sediment-laden runoff to enter the Mputi River.
Risk of sewer pipeline leakage	Leakage, rupture or failure of the sewer pipeline could result in sewage entering the Mputi River, potentially causing deterioration of water quality and associated impacts on aquatic and riparian habitat.
Ongoing disturbance of riparian habitat	Routine maintenance of the infrastructure may result in continued localised disturbance of riparian vegetation and associated habitat.

5.2 IMPACT ASSESSMENT

All identified impacts for the proposed development are assessed and findings are entailed in **Table 17**.

Table 17. Impact rating

	Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
Construction	Direct and Indirect											
	Non-compliance with environmental and water legislation	Without	1	3	3	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Loss/disturbance of riparian vegetation and associated habitat	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	2	2	5	Low	2	Probable	Low	-ve	High
	Disturbance and modification of the riverbed and banks	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	2	2	5	Low	2	Probable	Low	-ve	High
	Bank erosion and sedimentation	Without	1	2	3	6	Medium	3	Definite	Medium	- ve	High
		With	1	2	2	5	Low	2	Probable	Low	-ve	High
	Flow modification	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		With	2	2	1	5	Low	2	Probable	Low	- ve	High
	Deterioration of surface water quality	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
		With	2	2	1	5	Low	2	Probable	Low	-ve	High
	Cumulative Impacts											
	Cumulative impact on aquatic biodiversity	Without	2	3	2	7	High	3	Definite	High	- ve	High
		With	2	2	1	5	Low	2	Probable	Low	- ve	High
Operations	Direct and Indirect											
	Increased stormwater runoff	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
		Without	2	3	2	7	High	2	Probable	High	- ve	High

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	Risk of sewer pipeline leakage	With	1	2	1	4	Very low	2	Probable	Very Low	- ve	High
	Ongoing disturbance of riparian habitat	Without	1	2	2	5	Low	3	Definite	Low	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High

5.3 MITIGATION MEASURES

Table 18 identifies mitigation measures that have been identified during the assessment for the proposed development.

Table 18. Mitigation measures

Impact	Mitigation measures
Construction Phase	
Non-compliance with environmental and water legislation	Obtain all required environmental and water-related approvals prior to any site works.
Loss/disturbance of riparian vegetation and associated habitat	- Limit vegetation clearing to the area required for construction. - Clearly demarcate the approved working footprint and treat areas outside it as no-go areas. - Locate construction camps, laydown areas, and materials stockpiles outside of the riparian zone. - Use existing access routes where practicable. - Avoid unnecessary removal of indigenous riparian vegetation. - Rehabilitate disturbed areas immediately after construction.
Disturbance and modification of the riverbed and banks	- Excavation should be limited only to the approved footprint. - Restore disturbed banks after construction and stabilise exposed areas to prevent further disturbance.
Bank erosion and sedimentation	- Implement erosion and sediment-control measures. - Stabilise exposed soils and locate stockpiles away from the Mputi River, artificial dam/impoundment and associated drainage pathways.
Flow modification	- Works within the active channel should, where practicable, be undertaken during naturally low-flow conditions. - Disturbance within the channel must be restricted to the development footprint. - Temporary flow diversions, where required, must be removed after construction.
Deterioration of surface water quality	- Minimise works within the watercourses as far as practically possible. - Refueling and maintenance of machinery must take place away from the watercourse. - Mixed cement must not be allowed to flow into any watercourses. - Drip trays must be placed under stationary construction machinery overnight to avoid soil contamination from oil and fuel leaks - Spill-response kits must be available on site.
Cumulative impact on aquatic biodiversity	- Restrict all works to the construction footprint. - Where possible, existing disturbed areas should be used for access and working areas. - Rehabilitate disturbed areas immediately after construction.
Operation Phase	
Increased stormwater runoff	- Prevent direct discharge of concentrated runoff into the Mputi River. - Stormwater infrastructure must be regularly inspected and maintained.
Risk of sewer pipeline leakage	- Sewer pipeline must be designed and installed to minimise the risk of leakage at the river crossing. - Regular inspection and maintenance must be undertaken. - Any leak or failure must be repaired immediately.
Ongoing disturbance of riparian habitat	Avoid unnecessary clearing of riparian vegetation.

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	• Disturbed areas must be rehabilitated after maintenance activities, and alien invasive plants should be controlled where they establish.
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6. CONCLUSION

The aquatic biodiversity assessment for the proposed Dutywa Conference Centre indicates that the project area is associated with the Mputi River, a non-perennial watercourse that remains ecologically functional despite existing localised disturbance. The river and its associated riparian habitat provide locally important aquatic habitat and contribute to ecological connectivity within the broader river network and are therefore regarded as sensitive aquatic receptors within the project area.

Potential construction-related impacts include disturbance and loss of riparian habitat, modification of the riverbed and banks, erosion and sedimentation, temporary flow modification and potential deterioration of surface-water quality. Operational impacts are primarily associated with increased stormwater runoff, potential sewer pipeline leakage or failure and ongoing disturbance of riparian habitat. The nearby artificial dam may also be indirectly affected by sediment runoff generated during construction. The identified impacts are considered mitigable provided that construction activities are restricted to the development footprint and the recommended mitigation measures are implemented. Without mitigation, the identified impacts range predominantly from Medium to High significance. Following implementation of the recommended mitigation measures, the impacts are expected to reduce to Low or Very Low significance.

From an aquatic biodiversity perspective, the proposed development is therefore considered acceptable, provided that the recommended mitigation measures are effectively implemented and all required environmental and water related authorisations are compiled with.

7. REFERENCES

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